HACKATHON PROBLEM STATEMENT

DROPLET INTELLIGENCE: BUILDING PHYSICS-INFORMED AI MODELS FOR PREDICTING LIQUID–SURFACE INTERACTIONS

1. Problem Overview

The interaction between a liquid droplet and a solid surface plays a crucial role in applications such as anti-icing coatings, spray cooling systems, microfluidics, self-cleaning surfaces, and advanced manufacturing. When a droplet impacts an engineered surface, its behaviour can vary from rebound to spreading and permanent adhesion depending on fluid properties, impact conditions, and microscopic surface structures.

Traditional approaches use dimensionless parameters such as Reynolds Number (Re), Weber Number (We), and Ohnesorge Number (Oh) to describe droplet dynamics. However, complex micro-scale surface geometries remain difficult to capture using conventional correlations. Data-driven models can also struggle when they encounter surface conditions that were not represented during training.

2. Your Challenge

Develop a Physics-Informed Machine Learning (PIML) system that predicts droplet impact behaviour and generalizes effectively across different surface architectures. The solution should combine machine learning, fluid-dynamics principles, surface geometry information, and physics-based feature engineering.

3. Objectives

- **Droplet Behaviour Prediction:** Predict key impact characteristics such as maximum spreading behaviour, rebound tendency, wetting/pinning probability, or impact outcome classification.
- **Cross-Surface Generalization:** Evaluate performance on surface geometries that were not included during training.
- **Physics-Aware AI:** Explore dimensionless parameters (Re, We, Oh), physics-guided feature extraction, hybrid physics–AI architectures, and explainable ML.

4. Dataset Description

The research dataset contains droplet-impact experiments on laser-textured superhydrophobic aluminium alloy surfaces, with variations in surface geometry, droplet viscosity, droplet velocity, and droplet diameter. It also contains a smooth chemically identical reference surface that can be used to evaluate out-of-distribution generalization. The source research contains 1,498 textured-surface experiments and 125 reference-surface experiments.

5. Expected Deliverables

- **AI Prediction Model:** A trained model capable of predicting droplet impact behaviour.
- **Physics Integration Report:** Explanation of selected features, physical assumptions, model architecture, and training methodology.
- **Explainability Module:** Identification of physical parameters and surface characteristics that influence predictions.
- **Demonstration Interface:** A simple dashboard or API where users can enter droplet, impact, and surface parameters and obtain predictions.

6. Evaluation Parameters

Criterion	Description
Prediction Accuracy	Ability to accurately predict droplet behaviour.

Criterion	Description
Generalization Capability	Performance on unseen surface conditions.
Physics Integration	Effective incorporation of scientific and physical knowledge.
Explainability	Ability to interpret model decisions and influential parameters.
Innovation	Novel approaches beyond conventional machine-learning pipelines.

7. Final Goal

Create an AI system that can act as a virtual droplet laboratory—predicting how liquids interact with engineered surfaces before physical testing. The solution should demonstrate how Artificial Intelligence and Physics can work together to support next-generation surface design and engineering applications.

Theme: Physics-Informed Artificial Intelligence | Scientific Machine Learning | Fluid Dynamics | Surface Engineering | Explainable AI

Problem statement derived from the supplied research abstract, “Physics-Informed Machine Learning for Cross-Surface Generalization in Droplet Impact Prediction.”